

Explicit constants for the mixed remainder

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Abstract

Astra #9 R5, step 1. The mixed-remainder estimate was the only genuinely asymptotic step of the $d = 2$ cutoff theorem (an `IsBig0` with an existential constant). It is made explicit: `twoDGeneral`($\beta, a, b, N^2; h, k$) is the product kernel of the Gaussian kernel $g(s) = e^{-\beta s^2 + \beta a s}$, so the explicit product-kernel bounds of units 111–112 give, for $N \geq 1$, `twoDGeneral`(N^2) $\leq \text{mixConst} \cdot N^{-\min(p_1, p_2)} (1 + \log N)$ with `mixConst` a finite expression in the Gaussian moments $\int_0^\infty s^{p-1} g$ and $\int_0^\infty s^{p-1} \log_+(R/s) g$ (equal case) or a single moment (unequal case). Hence $|\text{twoDAmp } \Phi| \leq C e^{D^2/\beta} \text{mixConst}(\beta/2, 2L, b, h + M, k) N^{-E'} (1 + \log N)$ for an amplitude with the rectangular envelope (`twoDAmp_env_explicit`). Zero `sorry/axiom`.

1 The things formalised

```
theorem twoDGeneral_sq_eq_prodKernel ( a b N : ) (h h k k : ) (hN : 0 < N) :
  twoDGeneral a b (N ^ 2) h h k k = prodKernel b h h k k (quadKernel a) N
theorem quadKernel_moment_integrableOn ( a p : ) (h : 0 < ) (hp : 0 < p) :
  IntegrableOn (fun s => s ^ (p - 1) * quadKernel a s) (Ioi 0)
theorem logPlus_div_le (R s : ) (hR : 0 < R) (hs : 0 < s) :
  logPlus (R / s) (1 + |Real.log R|) * (1 + |Real.log s|)
theorem quadKernel_logMoment_integrableOn ( a p R : ) (h : 0 < ) (hp : 0 < p)
  (hR : 0 < R) :
  IntegrableOn (fun s => s ^ (p - 1) * logPlus (R / s) * quadKernel a s) (Ioi 0)
noncomputable def mixConst ( a b : ) (h h k k : ) :      -- Gaussian moments, by case
theorem twoDGeneral_sq_le ( a b : ) (h h k k : ) (h : 0 < ) (hb : 0 < b)
  (hk : 0 < k) (hk : 0 < k) (N : ) (hN : 1 < N) :
  twoDGeneral a b (N ^ 2) h h k k
  mixConst a b h h k k
  * N ^ (-min (((h : ) + 1) / k) (((h : ) + 1) / k)) * (1 + Real.log N)
theorem twoDAmp_env_explicit ( b L C N : ) (h h k k M M D : ) (Φ : → → → )
  (h : 0 < ) (hb : 0 < b) (hk : 0 < k) (hk : 0 < k) (hC : 0 < C) (hN : 1 < N)
  (hΦ : Continuous fun x : × × => Φ x.1 x.2.1 x.2.2)
  (hbound : u Icc (0 : ) b, v Icc (0 : ) b, s, 0 < s →
    |Φ u v s| < C * (u ^ M * v ^ M) * ((1 + s) ^ D * Real.exp ( * s * L))) :
  |twoDAmp b N h h k k Φ|
  C * Real.exp ((D : ) ^ 2 / ) * mixConst ( / 2) (2 * L) b (h + M) (h + M) k k
  * N ^ (-min (((h + M : ) : ) + 1) / k) (((h + M : ) : ) + 1) / k))
  * (1 + Real.log N)
```

2 Mathematics

With $t = N$ and $n = N^2$, $e^{-\beta n(u^k v^k)^2 + \beta \sqrt{n} u^k v^k a} = g(N u^k v^k)$, so the double integral is the product kernel $\iint u^{h_1} v^{h_2} g(N u^{k_1} v^{k_2})$. In the equal case $p_1 = p_2 = p$ the product-kernel bound reads

$N^{-p}(k_1 k_2)^{-1}(\log N M_p(g) + U_{p,R}(g)) \leq (M_p + U_{p,R})(k_1 k_2)^{-1} N^{-p}(1 + \log N)$; in the unequal case $p_1 < p_2$ it is $b^{k_2(p_2-p_1)} M_{p_1}(g)/(k_1 k_2(p_2 - p_1)) N^{-p_1}$ and the factor $1 + \log N \geq 1$ is added for a uniform shape; the case $p_2 < p_1$ follows by swapping the two variables. The moments exist because $g(s) = e^{-\beta s^2} e^{\beta a s}$ has the envelope form accepted by the moment-integrability lemma of unit 96, and $\log_+(R/s) \leq (1 + |\log R|)(1 + |\log s|)$ reduces the log-moment to the $j = 1$ case.

3 Paper-facing meaning

This is the first step of Astra #9's R5 (explicit constants, then uniform families): the remainder constant of the mixed term is now a named expression in Gaussian moments, as Astra recommended ("explicit enough, and maintainable"). What remains for a fully explicit $|Z(N) - S(N)| \leq K N^{-E}(1 + \log N)$: the face transfer constants (already explicit in `face_transfer_bound`, but stated through moments of the actual face coefficients, which must be bounded by envelope moments for uniformity), the corner transfer tails, and the discard step.

4 Lean notes

- To match $\iint u^h v^h e^{(\cdot)}$ with $\int u^h \int v^h g$, pull the outer factor into the inner integral with `rw [← integral_const_mul]` and identify the exponents with a `show ... by ring rewrite` rather than `congr` chains.
- `Measurable.div` with constants (`measurable_const.div measurable_id`) handles $s \mapsto R/s$; `logPlus` must be unfolded to `max` first.
- Case analysis on the starting exponents via `lt_trichotomy` with `if_pos/ if_neg` on the definition of `mixConst`; the third case rewrites through `twoDGeneral_swap` back and forth around `twoDGeneral_sq_eq_prodKernel`.
- `le_mul_of_one_le_right` adds the harmless factor $1 + \log N \geq 1$.